

# ASLR on the Line: Practical Cache Attacks on the MMU

**ASLR on the Line: Practical Cache Attacks on the MMU** - Author not stated, NDSS Symposium.

- Published: date not stated
- Original: <<https://www.ndss-symposium.org/ndss2017/ndss-2017-programme/aslrcache-practical-cache-attacks-mmu/>>
- Preserved from: <https://www.ndss-symposium.org/ndss2017/ndss-2017-programme/aslrcache-practical-cache-attacks-mmu/> (live) on 2026-08-08
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**\*\*Date:** **\*\*27 Feb 2017**

**\*\*Document Type:** **\*\*Reports**

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## Abstract:

Address space layout randomization (ASLR) is an important first line of defense against memory corruption attacks and a building block for many modern countermeasures. Existing attacks against ASLR rely on software vulnerabilities and/or on repeated (and detectable) memory probing.

In this paper, we show that neither is a hard requirement and that ASLR is fundamentally insecure on modern cachebased architectures, making ASLR and caching conflicting requirements (ASLR Cache, or simply AnC). To support this claim, we describe a new EVICT+TIME cache attack on the virtual address translation performed by the memory management unit (MMU) of modern processors. Our AnC attack relies on the property that the MMU's page-table walks result in caching page-table pages in the shared last-level cache (LLC). As a result, an attacker can derandomize virtual addresses of a victim's code and data by locating the cache lines that store the page-table entries used for address translation.

Relying only on basic memory accesses allows AnC to be implemented in JavaScript without any specific instructions or software features. We show our JavaScript implementation can break code and heap ASLR in two major browsers running on the latest Linux operating system with 28 bits of entropy in 150 seconds. We further verify that the AnC attack is applicable to every modern architecture that we tried, including Intel, ARM and AMD. Mitigating this attack without naively disabling caches is hard, since it targets the low-level operations of the MMU. We conclude that ASLR is fundamentally flawed in sandboxed environments such as JavaScript and future defenses should not rely on randomized virtual addresses as a building block.

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